

ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

LAB06: On-Device Learning

Davide Nadalini – d.nadalini@unibo.it

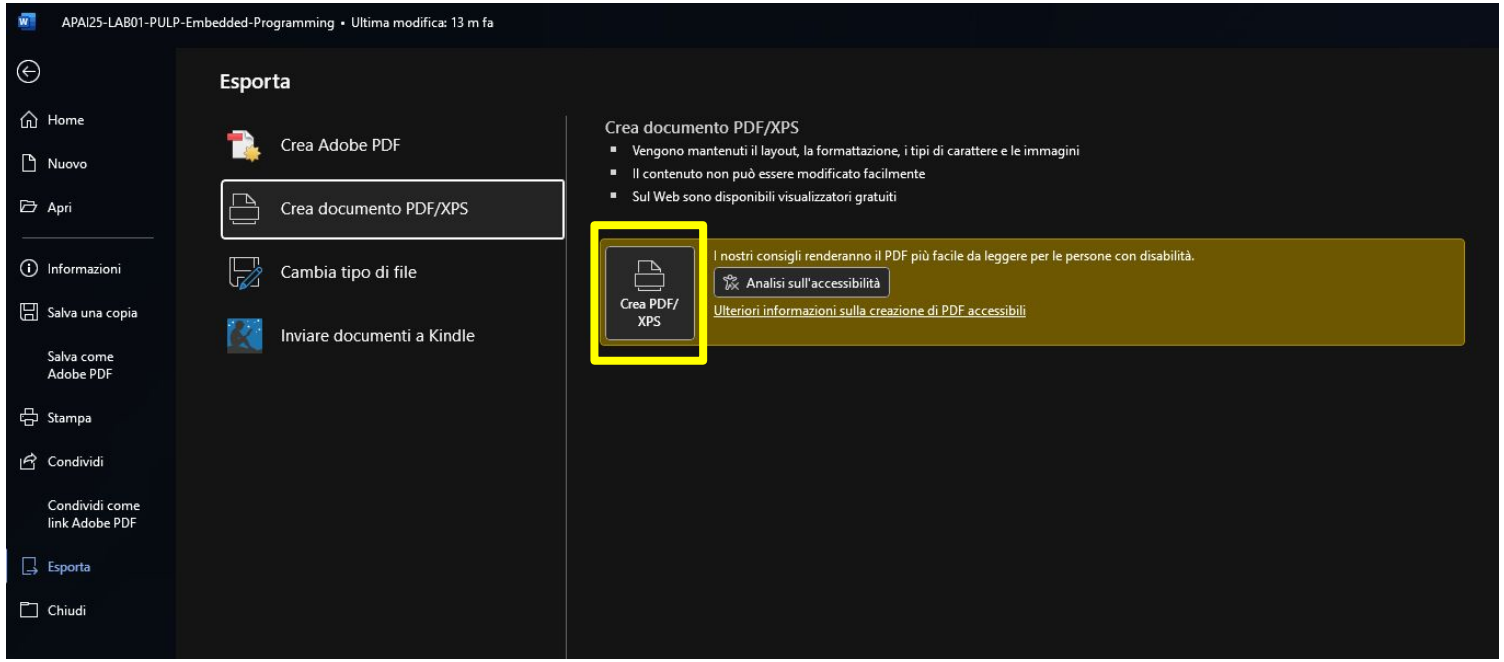
Alberto Dequino – alberto.dequino@unibo.it

Francesco Conti – f.conti@unibo.it

How to deliver the Assignment

You will deliver ONLY the PDF assignment, no code

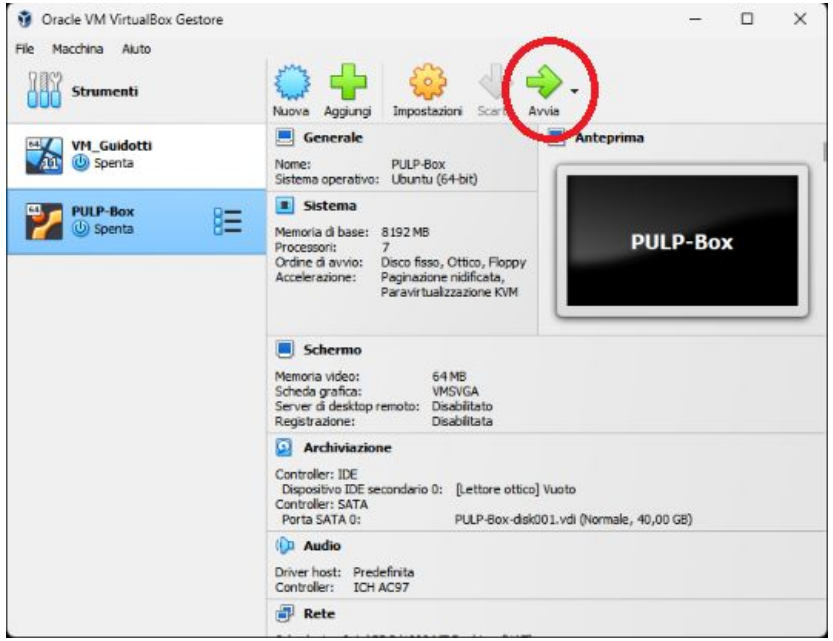
- Download the assignment file from Virtuale.
- Fill the results required by the assignment.
- Export to pdf format.
- Rename the file to: LAB<number_of_the_lesson>_APAI_<your_name>.pdf
- Use Virtuale platform to load ONLY your .pdf file

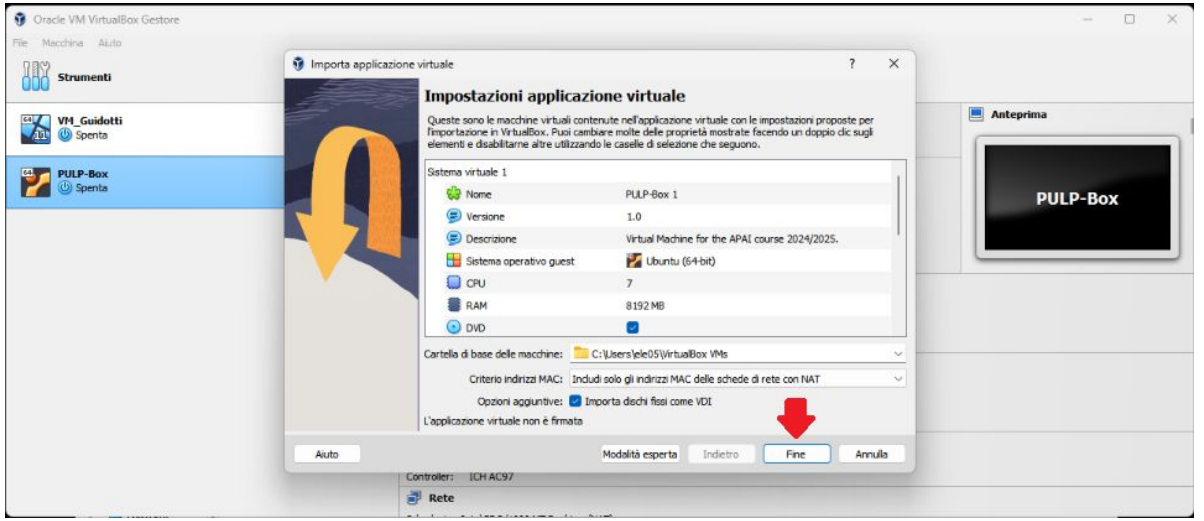


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Opening the VM and VSCode

1. On the lab's PCs, open the file explorer and go to This PC, C:/VM_APAI/
2. Double click on PULP-box.ova
3. VirtualBox opens, just click on "Fine"
4. Wait for the VM to be imported
5. Open the VM with "Avvia"





Password is 'pulp'

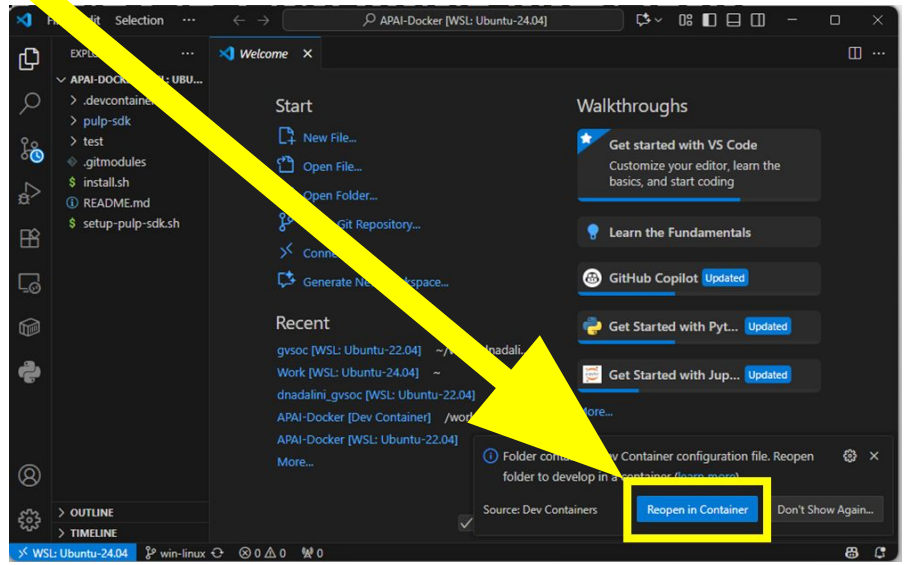
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Opening the Docker with VSCode

1. Open a terminal (right click – open a new terminal)
2. From the terminal, open VSCode in the folder of the Docker

```
$ cd APAI-Docker
$ code .
```

3. Reopen the APAI-Docker folder in VSCode (click on “Reopen in container”)
4. Now you can use the integrated terminal (open with CTRL+J) to run your applications!



IMPORTANT: every time you open a new terminal to work on PULP, launch

```
$ source setup-pulp-sdk.sh
```


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Getting Started:

IMPORTANT: activate the pulp-sdk module file every time a new shell is open.

```
$ source setup-pulp-sdk.sh
```

HOW TO RUN THE CODE:

```
$ git clone https://github.com/EEESlab/APAI25-LAB06-On-Device-Learning
$ cd APAI25-LAB06-On-Device-Learning
$ make clean get_golden all run
```



New rule today: this generates the python golden model!
The python script that is called is “utils/GM.py”

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On-Device Learning with PULP-TrainLib

Objective of the Class

Intro: On-Device Learning on MicroControllers

Tasks: Implement the C code to train a simple DNN on-device:

- Visualize tensors
- Forward, loss, backward, update: familiarize with the C code
- Implement a loss function on-device
- Train a DNN the GVSoC simulator with sample data and label

Deadline:

Nov 21nd 2025

Programming Language: C

Lab duration: 3h

<u>The class is meant to be interactive: coding together and on your own!</u>

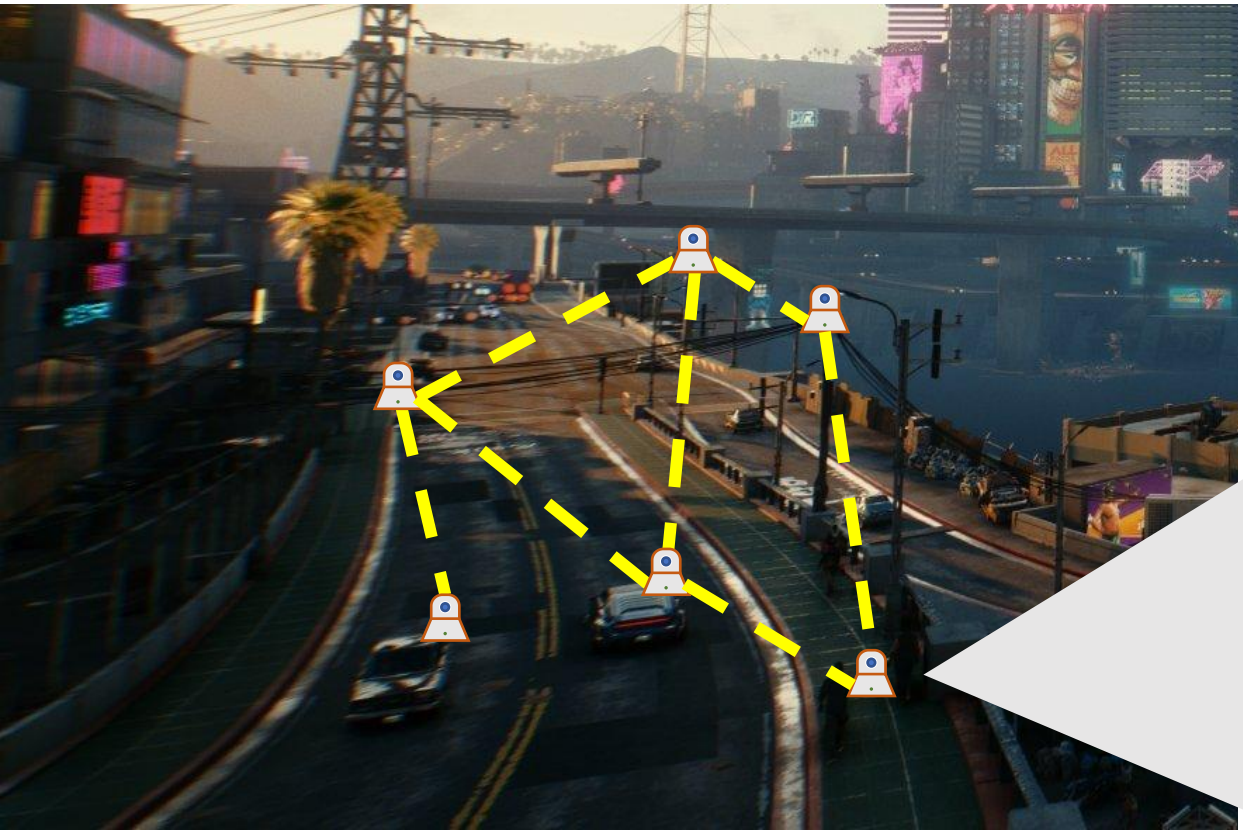
On-Device Learning (ODL)

"The process of **locally optimizing AI** models deployed on **Edge IoT Devices**"



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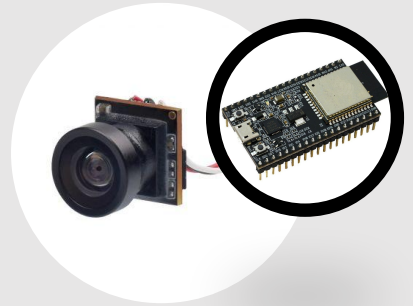
The Internet of Things (IoT)



IoT

"The process of connecting everyday physical objects to the internet"[1]

Miniaturized **low-power** and **low-cost edge** sensor nodes embedded in physical systems



[1]https://www.redhat.com/en/topics/internet-of-things/what-is-iot

IoT Applications – Examples

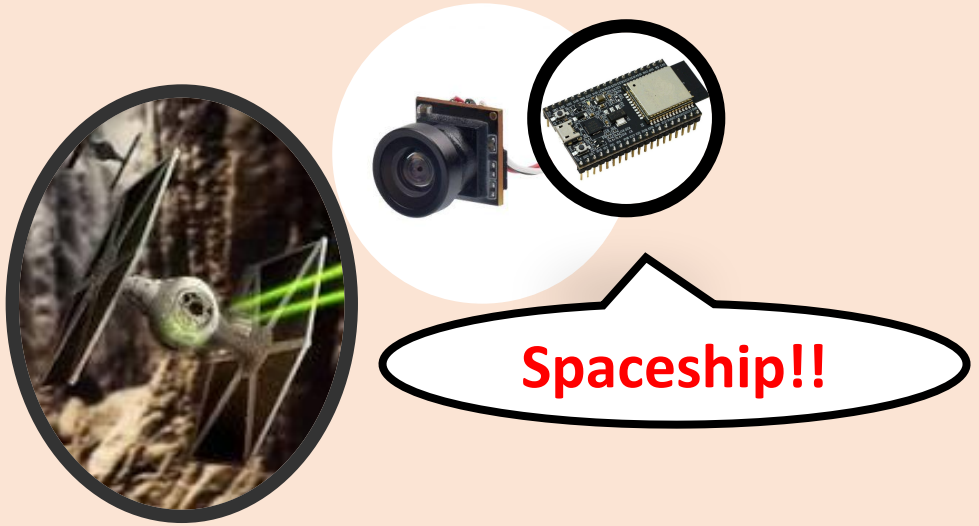


KEYWORD SPOTTING

Identify & React to user's keywords

IMAGE CLASSIFICATION

Detect the class of an image's content



IoT Applications – Examples (cont'd)

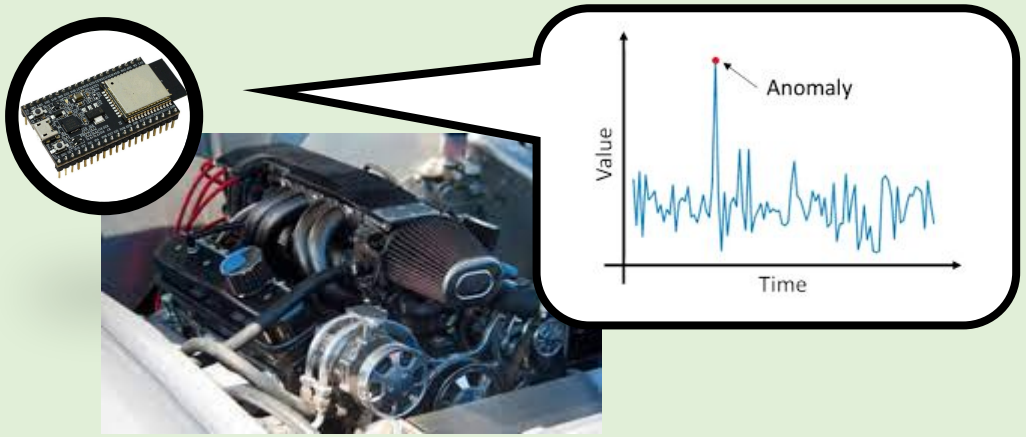


VISUAL WAKE WORDS

Detect the presence of a key object

ANOMALY DETECTION

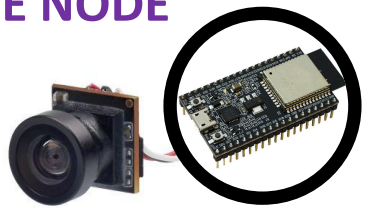
Detect when a signal indicates an anomaly



IoT Sensor Nodes: Tiny Machine Learning

LOW-POWER EDGE NODE





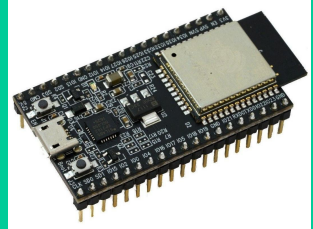
Spaceship!!

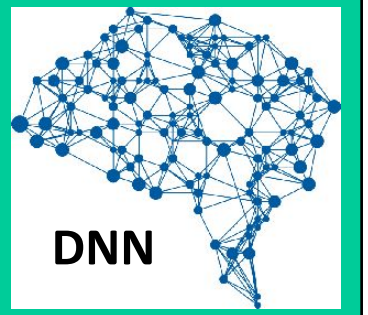
CAMERA & SENSORS

MCU
< 1 W
< 256 kB

A.I.







LOW-RESOLUTION CAMERAS AND SENSORS

ULTRA-LOW POWER

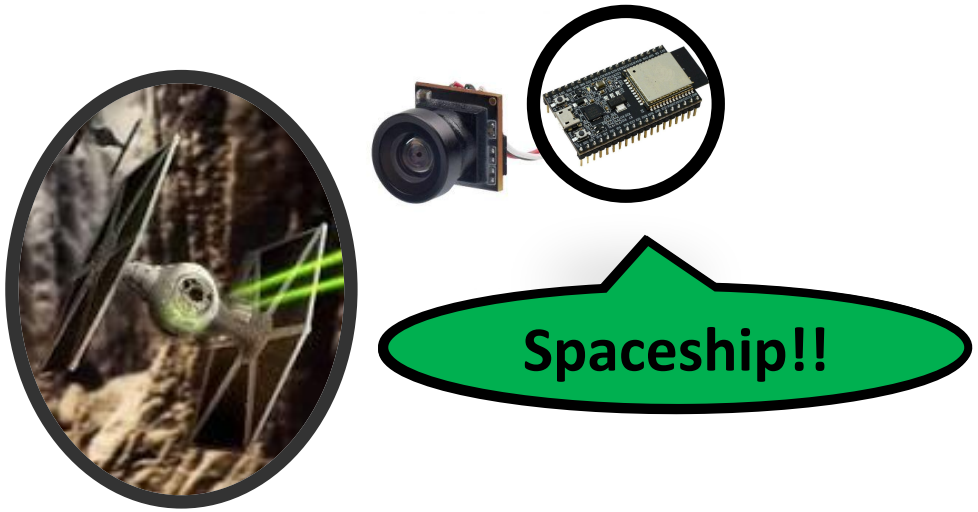
SMALL MEMORIES

LIMITED COMPUTATIONAL RESOURCES

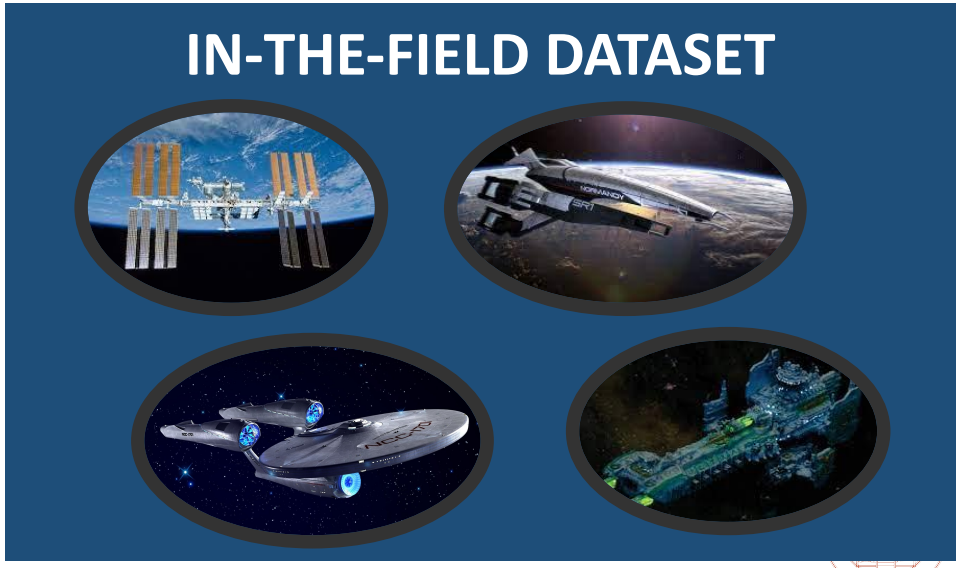
TINY MACHINE LEARNING

Reduce the AI requirements to fit tiny Devices

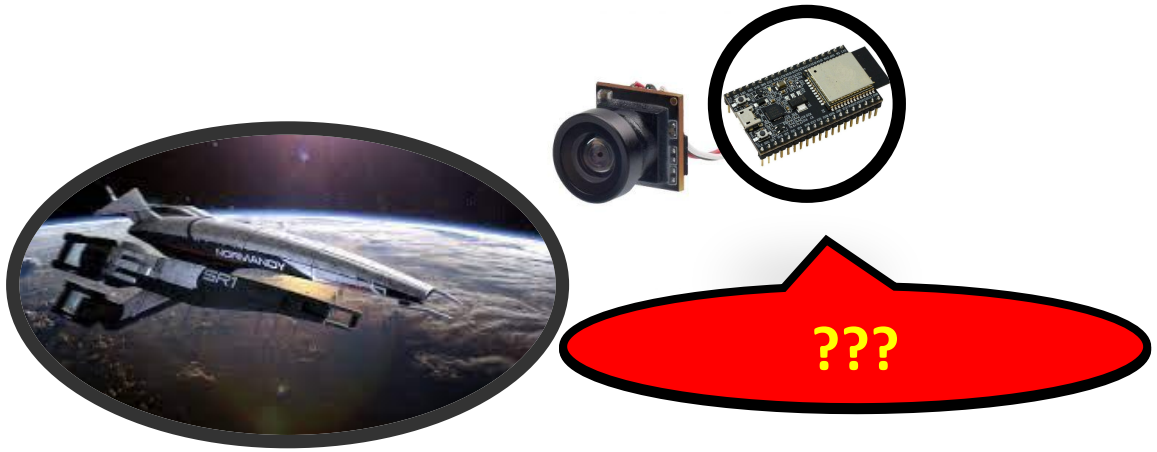
The inference problem



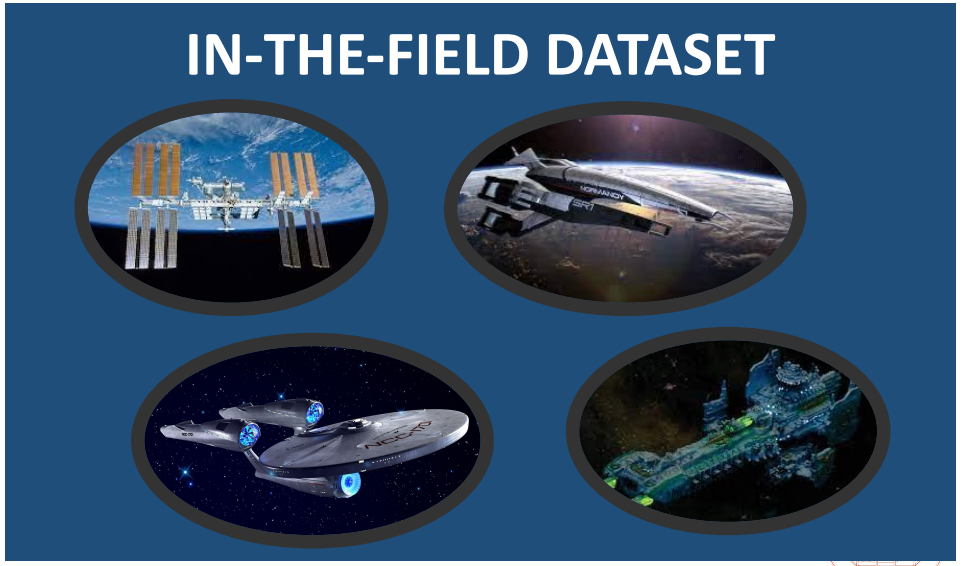




The inference problem





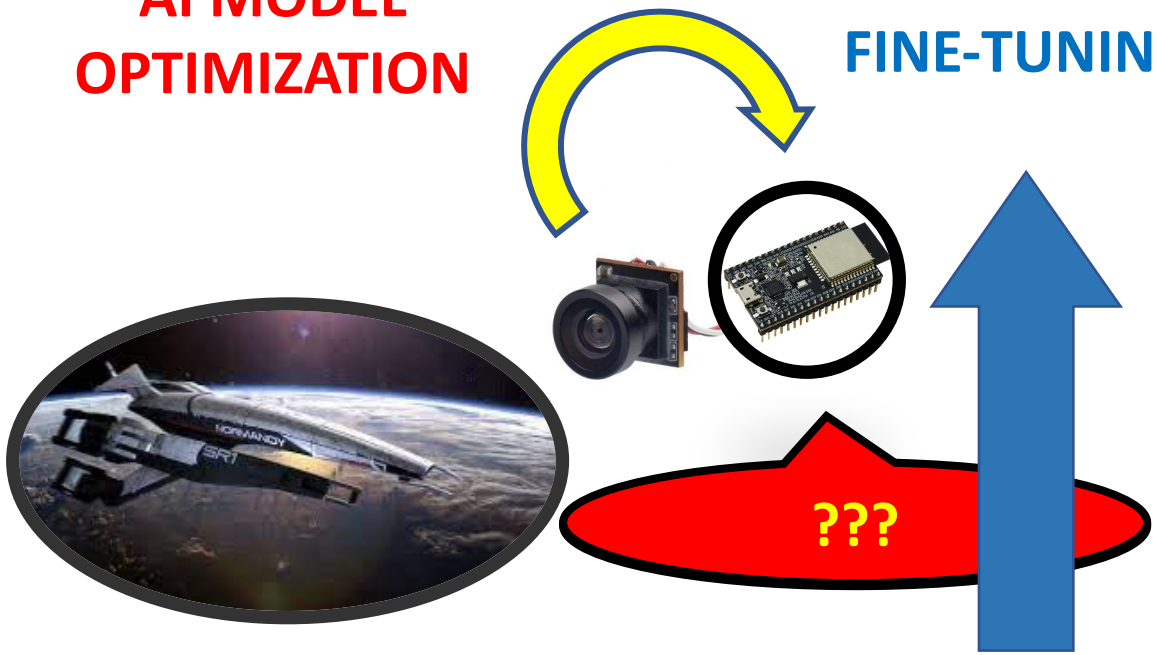


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The inference problem

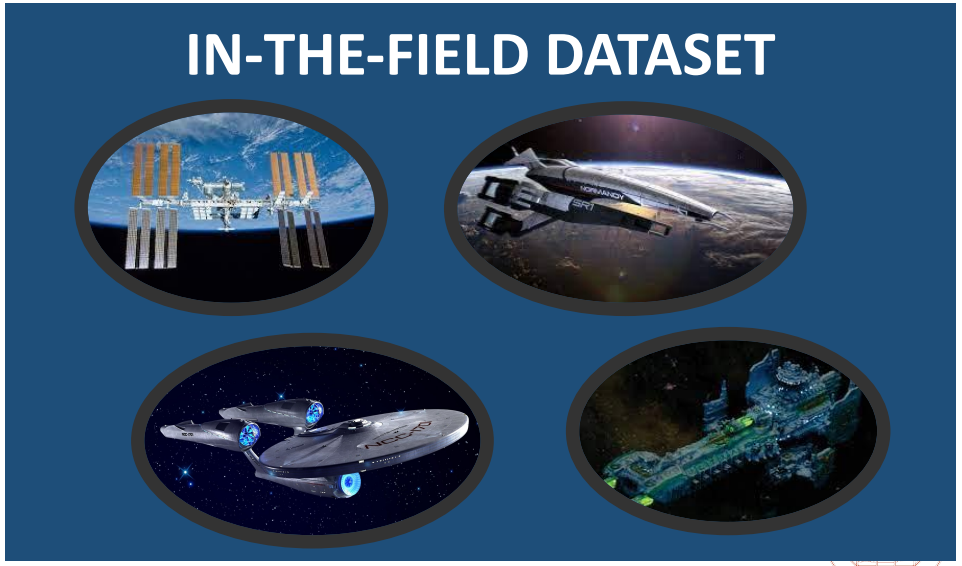
AI MODEL OPTIMIZATION

LOCAL RETRAINING / FINE-TUNING



On-Device Learning (ODL)

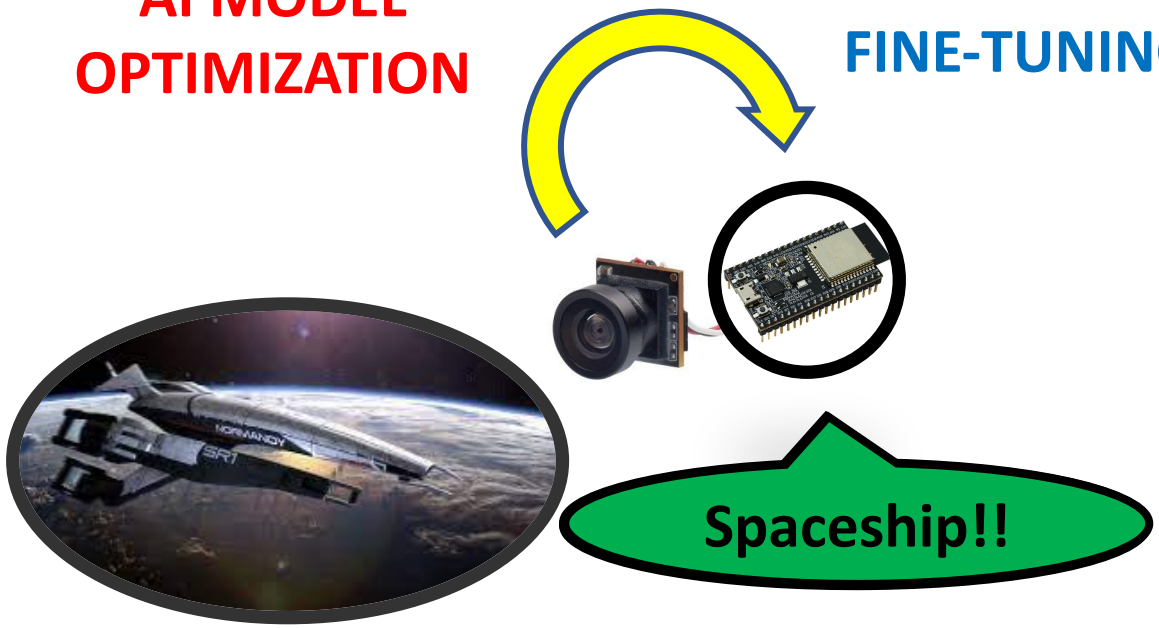




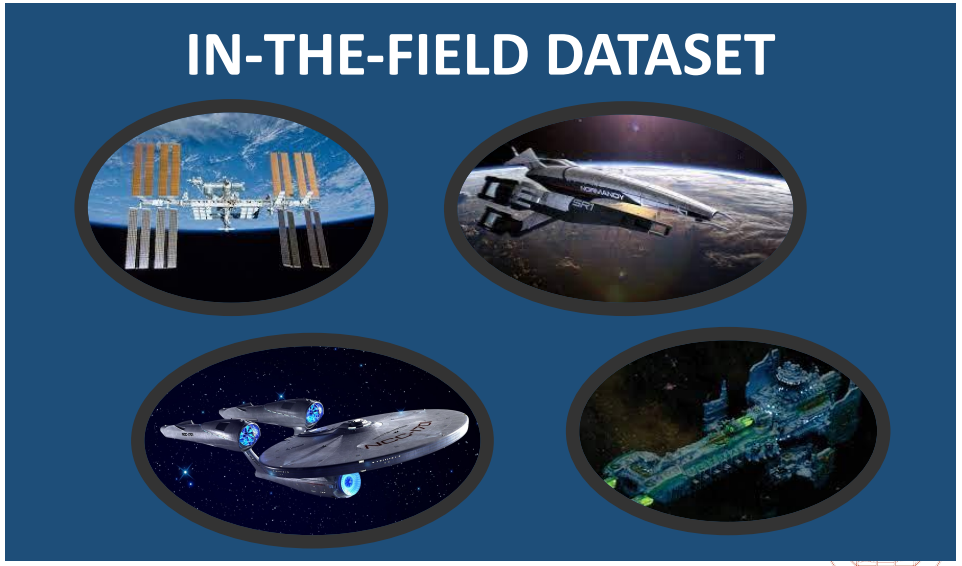
The inference problem

AI MODEL OPTIMIZATION

LOCAL RETRAINING / FINE-TUNING







On-Device Learning (ODL)

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On-Device Learning vs Training on Cloud

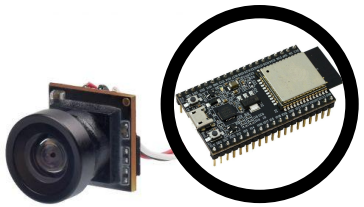
TRAINING ON CLOUD

CLOUD RETRAINING (full dataset)



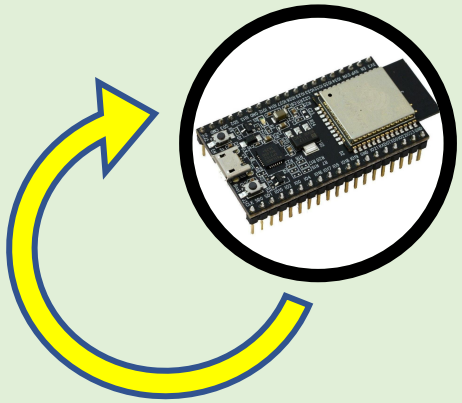
SEND NEW DATA

DEPLOY



???

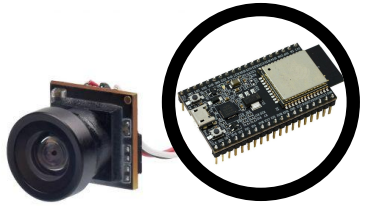
ON DEVICE LEARNING



LOCAL RETRAINING ON NEW DATA



ONLY NEW DATA: RISK OF CATASTROPHIC FORGETTING



???

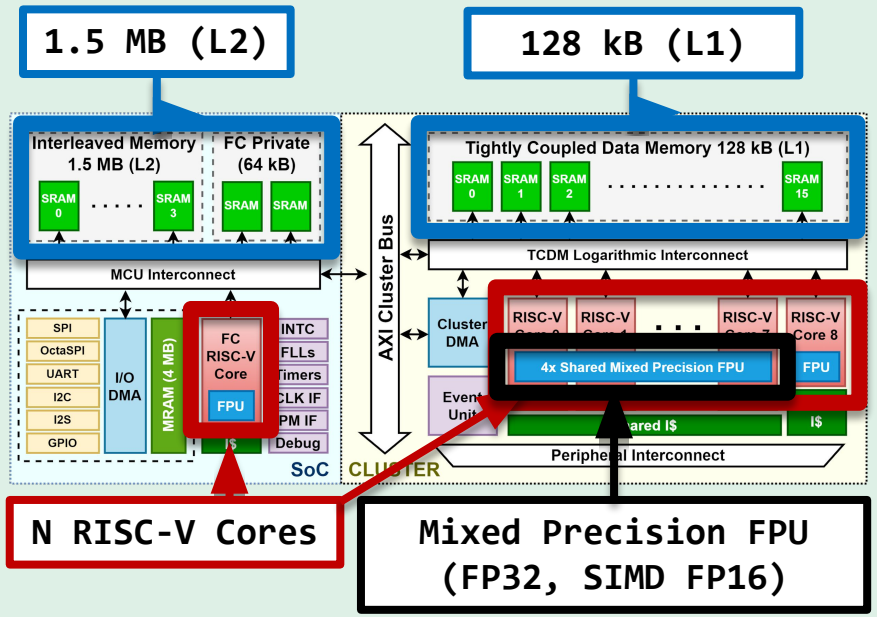
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ADVANTAGES OF On Device Learning

- ✓ Network Scalability
- ✓ User Data Privacy
- ✓ Latency of DNN Update

On-Device Learning on PULP

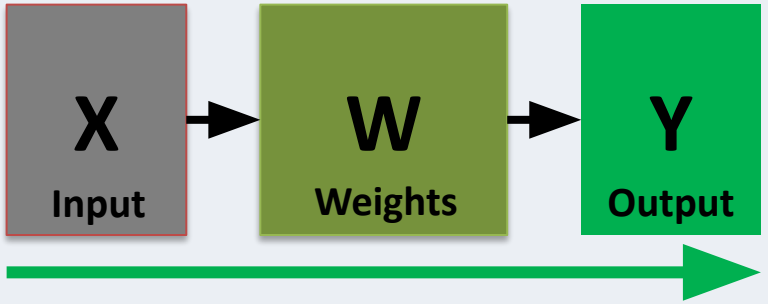
Our target: the PULP Platform[1]



PULP (Parallel Ultra-Low-Power)
Scalable, energy-efficient edge computing based on **RISC-V cores**.

Our task: The Backpropagation Algorithm

Step 1: Forward
Step 2: Loss
Step 3: Out grad



Forward: layer-wise predict the class / attribute of input feature map, then compute loss + output gradient

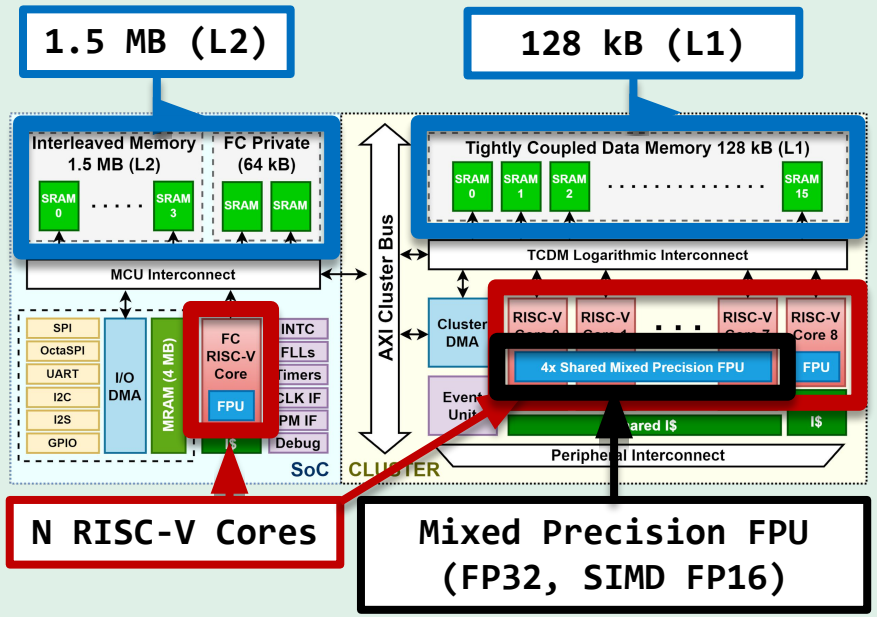


[1]PULP-Platform: https://pulp-platform.org/

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On-Device Learning on PULP

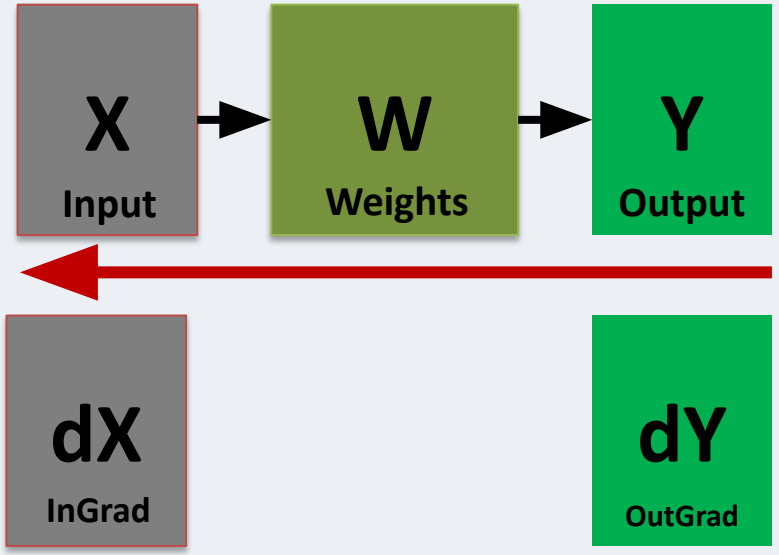
Our target: the PULP Platform[1]



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Step 4: Backward



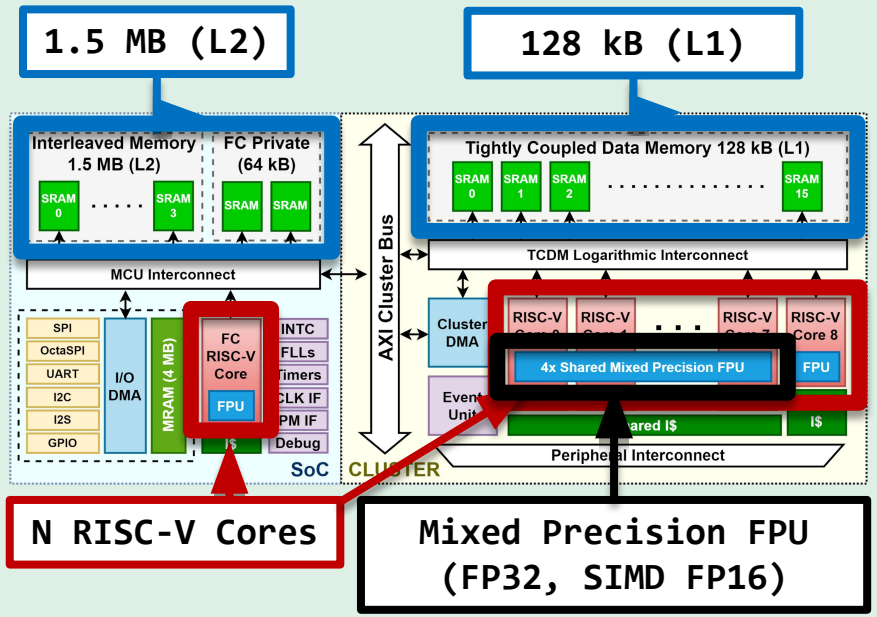
Step 1: Input Gradient Backpropagate the error dY towards previous layers

[1]PULP-Platform: https://pulp-platform.org/

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On-Device Learning on PULP

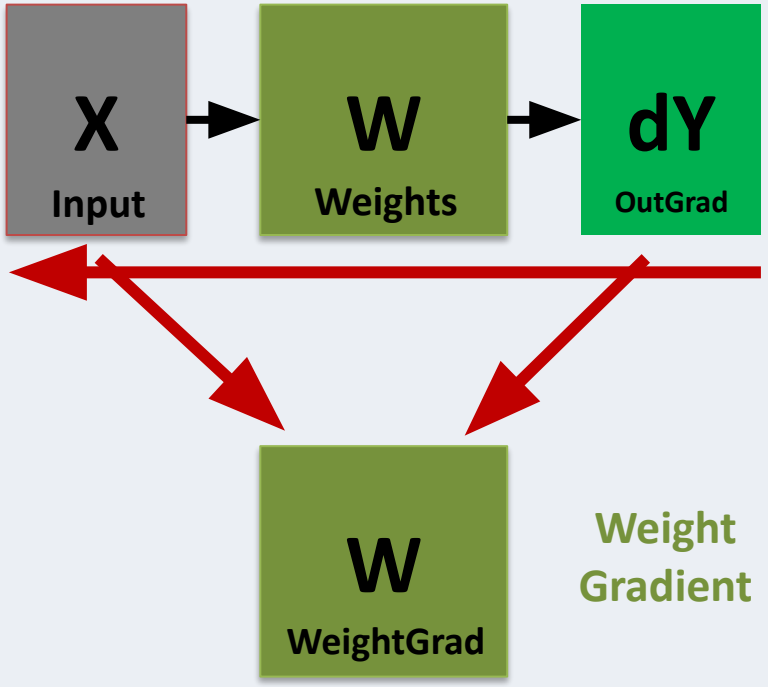
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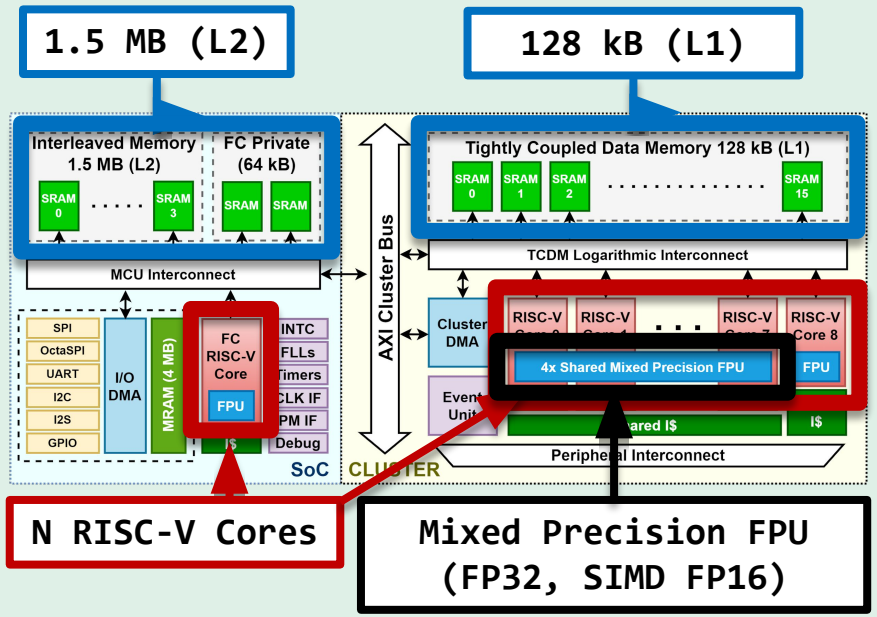
Step 2: Weight Gradient
Compute the optimization direction of weights

[1]PULP-Platform: https://pulp-platform.org/

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On-Device Learning on PULP

Our target: the PULP Platform[1]



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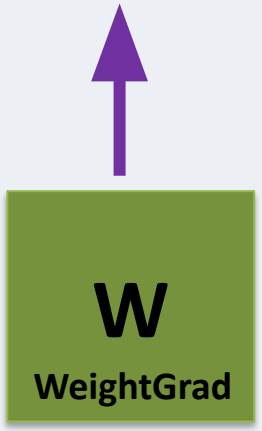
Our task: The Backpropagation Algorithm

Step 1: Forward
Step 2: Loss
Step 3: Out grad
Step 4: Backward
Step 5: Weight Update



OUR GOAL
Enhancing DNNs On-Device

Update with Learning Rate LR

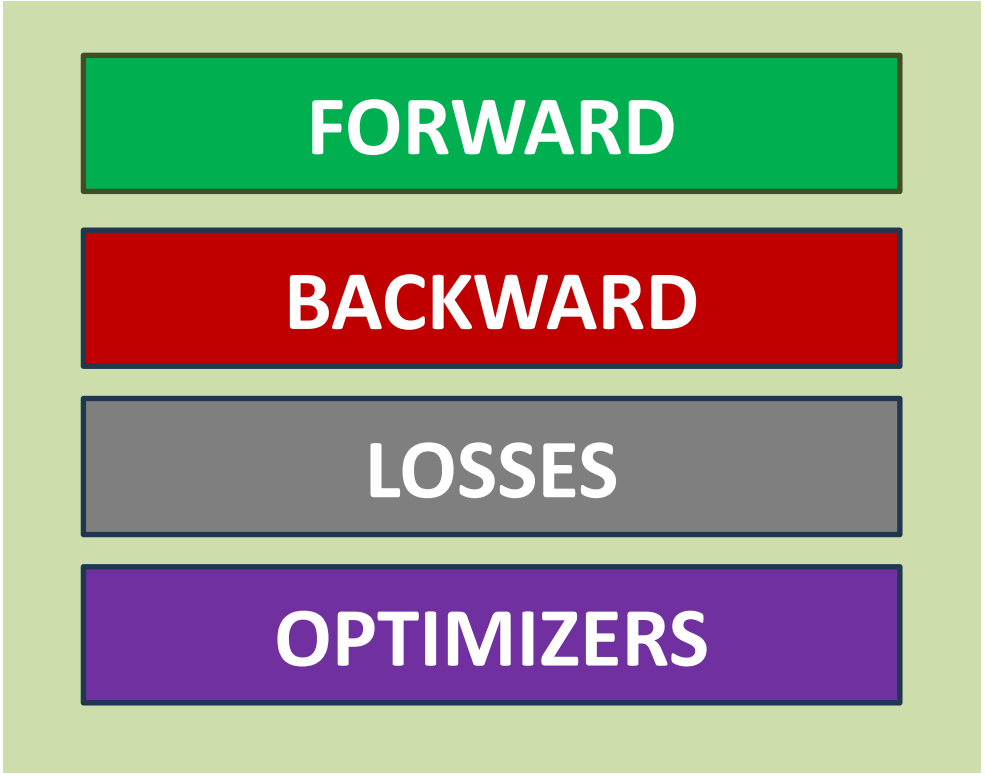


[1]PULP-Platform: https://pulp-platform.org/

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PULP-TrainLib

The first On-Device Learning library for RISC-V MultiCore MCUs (PULP)



E.G: Conv2D Training

```
// Arguments
struct Conv2D_args C2D_args;
struct loss_args loss_args;
struct optim_args optim_args;

// Forward layer
pulp_conv2d_fp32_fw_cl(&C2D_args);

// Loss function
pulp_CrossEntropyLoss(&loss_args);

// Backward
pulp_conv2d_fp32_bw_input_grads_cl(&C2D_args);
pulp_conv2d_fp32_bw_param_grads_cl(&C2D_args);

// Update
pulp_gradient_descent_fp32(&optim_args);
```

https://github.com/pulp-platform/pulp-trainlib

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Thank you for your attention

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Francesco Conti – f.conti@unibo.it